Practical: 08

Aim: A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Optimal page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3).

CODE:

```
~

user@AnushkaGudadhe MSYS ~
$ nano opt.c
```

```
~
  GNU nano 8.7
#include <stdio.h>

int main() {
    int pages[] = {1,2,3,4,1,2,5,1,2,3,4,5};
    int frames[3];
    int n = 12, frames_count = 3;
    int i, j, k, pos, farthest;
    int hit = 0, fault = 0;
    int flag;

    for(i = 0; i < frames_count; i++)
        frames[i] = -1;

    for(i = 0; i < n; i++) {
        flag = 0;

        // Check for page hit
        for(j = 0; j < frames_count; j++) {
            if(frames[j] == pages[i]) {
                hit++;
                flag = 1;
                break;
            }
        }

        // Page fault
        if(flag == 0) {
            pos = -1;
            farthest = i + 1;

            for(j = 0; j < frames_count; j++) {
                int found = 0;
                for(k = i + 1; k < n; k++) {
                    if(frames[j] == pages[k]) {
                        if(k > farthest) {
                            farthest = k;
                            pos = j;
                        }
                        found = 1;
                        break;
                    }
                }
```

```
            if(pos == -1)
                pos = 0;

            frames[pos] = pages[i];
            fault++;
        }
    }

    float hit_percent = (hit * 100.0) / n;
    float fault_percent = (fault * 100.0) / n;

    printf("Total Page Hits: %d\n", hit);
    printf("Total Page Faults: %d\n", fault);
    printf("Page Hit Percentage: %.2f%%\n", hit_percent);
    printf("Page Fault Percentage: %.2f%%\n", fault_percent);

    return 0;
}

^G Help        ^O Write Out   ^F Where Is    ^K Cut         ^T Execute     ^C Location    M-U Undo
^X Exit        ^R Read File   ^\ Replace     ^U Paste       ^J Justify     ^/ Go To Line  M-E Redo
```

OUTPUT:

```
user@AnushkaGudadhe MSYS ~
$ gcc opt.c -o opt

user@AnushkaGudadhe MSYS ~
$ ./opt
Total Page Hits: 5
Total Page Faults: 7
Page Hit Percentage: 41.67%
Page Fault Percentage: 58.33%

user@AnushkaGudadhe MSYS ~
$ |
```